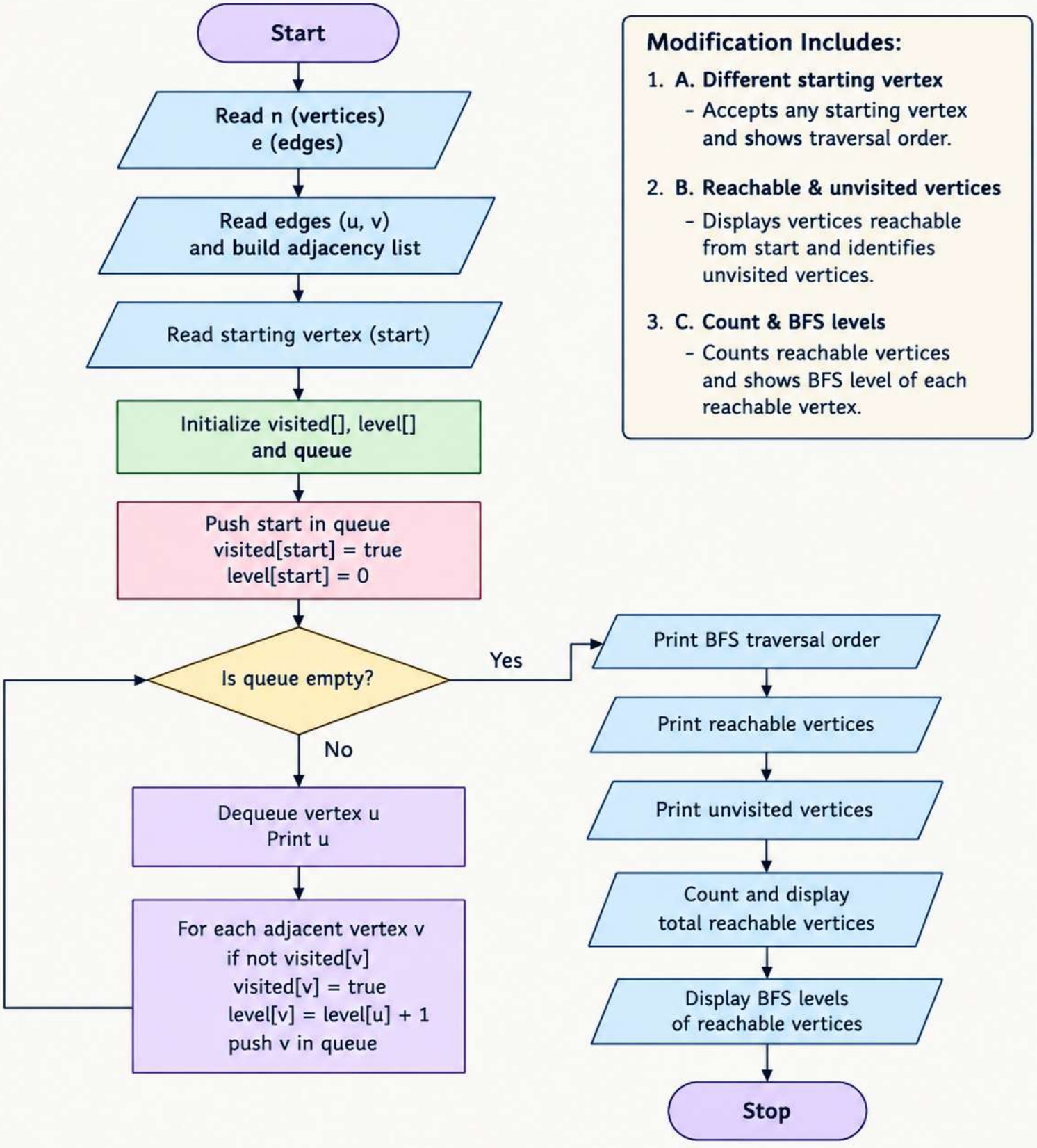

Start
Read n (vertices)
e (edges)
Read edges (u, v)
and build adjacency list
Read starting vertex (start)
Initialize visited[], level[]
and queue
Push start in queue
visited[start] = true
level[start] = 0
Is queue empty?
Yes
No
Dequeue vertex u
Print u
For each adjacent vertex v
if not visited[v]
visited[v] = true
level[v] = level[u] + 1
push v in queue
Print BFS traversal order
Print reachable vertices
Print unvisited vertices
Count and display
total reachable vertices
Display BFS levels
of reachable vertices
Stop
Modification Includes:
1. A. Different starting vertex
- Accepts any starting vertex and shows traversal order.
2. B. Reachable & unvisited vertices
- Displays vertices reachable from start and identifies unvisited vertices.
3. C. Count & BFS levels
- Counts reachable vertices and shows BFS level of each reachable vertex.